

[BUG] QNode with both ``expval`` and ``var`` measurements returns wrong re

<https://github.com/PennyLaneAI/pennylane/issues/4416>

ROW 32

[BUG] QNode with both `expval` and `var` measurements returns wrong results #4416

New issue

Closed

#4426



edddddy opened on Jul 31, 2023

Expected behavior

Consider the below code:

```
dev = qml.device('default.qubit', wires=2)

@qml.qnode(dev)
def circuit1():
    qml.Hadamard(0)
    qml.Hadamard(1)
    return qml.expval(qml.PauliX(0)), qml.expval(qml.PauliZ(1)), qml.var(qml.PauliZ(0))

@qml.qnode(dev)
def circuit2():
    qml.Hadamard(0)
    qml.Hadamard(1)
    return qml.expval(qml.PauliX(0)), qml.var(qml.PauliZ(0)), qml.expval(qml.PauliZ(1))
```

Executing `circuit1` should return

```
(tensor(1., requires_grad=True), tensor(0., requires_grad=True), tensor(1., requires_grad=True))
```

and executing `circuit2` should return

```
(tensor(1., requires_grad=True), tensor(1., requires_grad=True), tensor(0., requires_grad=True))
```

Actual behavior

Executing `circuit1` returns

```
(tensor(1., requires_grad=True), tensor(0., requires_grad=True), tensor(0., requires_grad=True))
```

and executing `circuit2` returns

```
(tensor(1., requires_grad=True), tensor(0., requires_grad=True), tensor(1., requires_grad=True))
```

Additional information

No response

Source code

No response

Tracebacks

No response

System information

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fix 'split_non_commuting' when tape contains both 'expval' and 'var' measurements
PennyLaneAl/pennylane

Notifications

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Participants



System information

```
Name: PennyLane
Version: 0.32.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/PennyLaneAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: c:\users\edward.jiang\appdata\local\programs\python\python310\lib\site-packages
Editable project location: C:\Users\edward.jiang\Documents\pennylane
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
ntic-version, toml
Required-by: PennyLane-Cirq, PennyLane-Lightning
```

```
Platform info:      Windows-10-10.0.19045-SP0
Python version:     3.10.11
Numpy version:      1.23.5
Scipy version:      1.10.1
Installed devices:
- default.gaussian (PennyLane-0.32.0.dev0)
- default.mixed (PennyLane-0.32.0.dev0)
- default.qubit (PennyLane-0.32.0.dev0)
- default.qubit.autograd (PennyLane-0.32.0.dev0)
- default.qubit.jax (PennyLane-0.32.0.dev0)
- default.qubit.tf (PennyLane-0.32.0.dev0)
- default.qubit.torch (PennyLane-0.32.0.dev0)
- default.qutrit (PennyLane-0.32.0.dev0)
- null.qubit (PennyLane-0.32.0.dev0)
- cirq.mixedsimulator (PennyLane-Cirq-0.29.0)
- cirq.pasqal (PennyLane-Cirq-0.29.0)
- cirq.qsim (PennyLane-Cirq-0.29.0)
- cirq.qsimh (PennyLane-Cirq-0.29.0)
- cirq.simulator (PennyLane-Cirq-0.29.0)
- lightning.qubit (PennyLane-Lightning-0.31.0)
```

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



 edddddy added **bug** on Jul 31, 2023



 edddddy mentioned this on Aug 1, 2023

 [Fix split_non_commuting when tape contains both expval and var measurements #4426](#)



 edddddy closed this as **completed** in [#4426](#) on Aug 2, 2023

Fix `split_non_commuting` when tape contains both `expval` and `var` measurements #4426

<> Code

Merged edddddy merged 4 commits into `master` from `fix_non_commuting` on Aug 2, 2023

Conversation 6 Commits 4 Checks 0 Files changed 3 +36 -2



edddddy commented on Aug 1, 2023

Contributor

Context:

Executing a quantum tape containing both `expval` and `var` measurements would currently produce incorrect results. See the linked issue for more details.

Description of the Change:

Change `split_non_commuting` to iterate over the correct set of measurements.

Related GitHub Issues:

Fixes #4416



fix split non commuting

3e12408



github-actions bot commented on Aug 1, 2023

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



edddddy requested a review from a team 2 years ago

changelog

e34bed2

edddddy force-pushed the `fix_non_commuting` branch from `e7f981d` to `e34bed2` 2 years ago

Compare

Reviewers

rmoyard



timmysilv



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[\[BUG\] QNode with both `expval` and `var` m...](#)

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